

MAT 310, Handout 10: Poisson Distribution (3.8)

Motivating example: We want to know how many cars pass a given intersection in a fixed amount of time. Assume all cars are independent.

Idea: Divide the total time into n subintervals such that the following assumptions are true:

- $P(0 \text{ cars in subinterval}) =$
- $P(1 \text{ car in subinterval}) =$
- $P(2 \text{ or more cars in subinterval}) =$

Poisson RV: Let $\lambda > 0$. For $Y \sim \text{Poisson}(\lambda)$, the probability distribution is given by:

$$p(y) =$$

for $y = 0, 1, 2, \dots$

Note: On a TI-83/84, `poissonpdf( $\lambda, y$ )` gives $P(Y = y)$ and `poissoncdf( $\lambda, y$ )` gives $P(Y \leq y)$. These functions can be found by clicking “2nd” $\rightarrow$ “VARS” and then scrolling down.

Remark: Poisson RVs are typically used to model the count of (often rare) occurrences in a specified time interval. A common way to encounter a Poisson RV is through a *Poisson process*—a model that appropriately describes situations like our motivating example.

IMPORTANT: Suppose a Poisson process has mean number of occurrences *per unit* (e.g., of time) equal to λ . Then, the number of occurrences in a units has a Poisson distribution with mean equal to $a\lambda$.

Example 1. Suppose an average of 10 customers arrive at a bank each hour according to a Poisson distribution. Find the probability that:

(a) exactly 8 customers arrive at a bank between 9 and 10 am.

(b) at least 2 arrive between 2 and 3 pm.

(c) at least 16 arrive between 1:30 and 3:30 pm.

Example 2. For $Y \sim \text{Poisson}(\lambda)$, show that $\sum_y p(y) = 1$.

Expected Value and Variance of $Y \sim \text{Poisson}(\lambda)$

$$E(Y) = \quad \quad \quad V(Y) =$$

Proof (of expected value):